

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think> in the context of human B cells, CD21 is involved in the interaction with EBV, which is the pathogen that would be most affected by the modification of B cells to not express CD21 </think><answer> Epstein-Barr virus (EBV)</answer>

